



Our brains are actually bigger

Neuroscientists have discovered the endostiform nucleus which was previously an undiscovered part of the human brain. <https://digit.in/dec18-69-1>



Robots get sense of touch

Stanford University researchers have created an electronic glove that will impart the sense of touch upon robots. <https://digit.in/dec18-69-2>



Shown here are the language groups around the world.

Even within a minor subset of the languages spoken around the world, the differences in perception is quite evident

coloured shape - or isn't. Colour has been at the centrestage of this linguistic debate because colour is something that is experienced universally and also something that is encoded into humans at a biological level. Light forms a continuous spectrum of colour for anyone perceiving it. However, it is how we divide this spectrum into smaller colour portions that determines how we perceive the world. Red or green are not independent colours really, they are part of a continuous spectrum, which goes beyond the colours that we can see. If you see a rainbow, it extends to the ultraviolet and infrared as well, you just cannot see it. The researchers investigated exactly how different languages break up this spectrum, each of which is a named hue in that particular language. Apart from the hue, colours can also be dark or light.

As a part of this test, the researchers took Russian, German and Greek users to identify the differences and consistencies between languages for colour perception. There are a few deviant examples of the what makes this an interesting lot. For instance, both Greek and Russian have a native word for dark and light blue, but neither have an overarching word for blue, which is present in German. All three

have a category word for green. Why is this significant? Earlier research has shown that having a category word for anything - an object, a colour, or other experiences - leads to faster cognitive perception of items from that category. For instance, prior studies have shown that Russian speakers are better at perceiving light and dark blue as compared to English speakers. Maier and Rahman, the researchers of the paper in question, decided to take things one step further and check if this distinctive behaviour could cause people to not perceive something entirely.

THE EXPERIMENT

During the study, the participants were asked to look for a grey semi-circle in a series of images. This was simply to grab their attention, where the actual objective was to understand their reaction to a series of images that contained geometric shapes of a certain colour against backgrounds of different colours. The real critical test was about them spotting a coloured triangle and whether they spotting it varied with the changes in colour - the changes being various combinations of light and dark blue and green.

What they observed was the Greek speakers and Russian speakers were more likely to perceive a light blue tri-

angle against a dark blue background as compared to the green counterpart, whereas no such difference was observed in German speakers.

Rather than depending on sheer verbal feedback, Rahman and Maier also checked EEG reports for tests conducted during the study to compare the brain activity of the participants. The EEG data was in line with the perception difference that was observed between the speakers from the various language groups. The results of the test held at a very primal level.

While this doesn't prove conclusively that there's a deterministic link between language and colour perception, it seems to strongly point towards the fact that having a particular language as your native language or mother tongue does influence an involuntary, subconscious stage of colour perception and processing in the human brain. [4](#)

Source:

- Psychological Science

Recommended reading:

- Language, Thought, and Reality: Selected Writings of Benjamin Lee Whorf, MIT Press
- The Great Whorf Hypothesis Hoax by Dan Moonhawk Alford
- How does our language shape the way we think? By Lera Boroditsky